Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/02/2010
PI Name:	Dziegielewski, Jaroslaw Ph.D.		
Project Title:	Effects of prolonged exposure to space radiation on carcinogenesis and neuroendocrine differentiation in human prostate models		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Charlottesville	**State:**	VA
Zip Code:	22908-0383	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	01/01/2010	**End Date:**	12/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Period of performance changed to 01/01/2010-12/31/2012 (from 11/15/09-11/14/12) per NSSC info (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Larner, James (University of Virginia) Parsons, Sarah (University of Virginia Health System)		
Grant/Contract No.:	NNX10AC13G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The deleterious effects of radiation encountered in space presents a significant obstacle in our quest to explore the space. Unfortunately, due to the significant differences between space radiation and radiation encountered on Earth, the risks of cancer development following exposure to space radiation cannot be reliably predicted based solely on epidemiological data collected on Earth. This project will study the effects of space radiation on development and/or progression of prostate cancer. Prostate cancer is the most common cancer in men in the United States and the second leading cause of cancer deaths. It is also the most often diagnosed type of cancer in male astronauts. However, the connection between exposure to space radiation and the development of prostate cancer is not yet known. Several lines of evidence suggest that conversion of normal prostate or prostate cancer cells into cells secreting cell growth hormones, so called neuroendocrine cells, stimulate cancer development and/or progression. Strikingly, such conversion was recently reported in prostate cells exposed to prolonged irradiation. The low doses and long exposure times used in that study are not dissimilar to the doses and exposure times encountered during space travel. Therefore, we postulate that space radiation could increase the risk of prostate cancer development through the transformation of prostate cells into neuroendocrine cells during space flights. Since neuroendocrine cells seem to play an important role in the development of other important and common types of solid tumors, such as breast, lung or colon, the insight gained during this study should be also applicable to these cancers. The results obtained during the proposed studies will allow NASA to better estimate the prostate cancer health risk(s) of space exploration, as well as provide clues that will allow the development of novel preventative measures to protect astronauts.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	It is believed that space radiation can increase the risk of tumor development and progression. We hypothesize that space radiation could not only induce neoplastic transformation in prostate cells (carcinogenesis), but in addition to direct interactions with target cells (e.g. causing carcinogenic mutations in the cellular DNA), could also stimulate surrounding cells and tissues to provide growth assistance to the developing tumor. One of the mechanisms of such stimulation could be the neuroendocrine differentiation, converting normal epithelial cells into neuroendocrine-like cells which produce hormones and growth factors. Such differentiation has been observed in prostate, breast, GI tract and lung tissues, all important target organs for space radiation-induced cancers. Some of the research and technological questions that this proposal addresses include the following: 1) What is the relationship between induction of carcinogenesis and/or tumor progression and the dose and quality of space radiation? 2) Can the in vitro cell culture models be used as an approximation for prostate carcinogenesis and neuroendocrine differentiation in humans? 3) How can effective biomarkers of carcinogenic and neuroendocrine differentiation risk from space radiation be developed and validated? 4) What is the likelihood that neuroendocrine differentiation will increase the risk of carcinogenesis in prostate and other target organs? Better understanding of the mechanisms underlying the effects of exposure to radiation, including the role of stimulating activities of cancer and surrounding cells, will provide significant advantages in predicting cancer risk from space and/or terrestrial radiations.		

Task Progress:	During the first year of this project, we received two hours of proton (NSRL10B, LET~0.25 keV/n) and two hours of oxygen ions (NSRL10C, LET~25 keV/n) irradiation time. These allowed us, together with X-ray irradiation experiments at the University of Virginia, to begin establishing relationship between the radiation dose / quality and the biological responses in irradiated prostate cells. Due to the time limitation, only the relatively short-term assessments have been made so far; however, the cell stocks from both of these runs will be grown and evaluated for radiation-induced neoplastic transformation and neuroendocrine differentiation following longer time periods. We are approved for additional beamtime during NSRL11A (Spring 2011) to irradiate cells with titanium ions (1000 MeV/n, LET ~100 keV/n) and complete planned ranges of LET dependence. Our preliminary studies suggest that cellular responses to ionizing radiation will depend on radiation quality (e.g. LET) and dose, and also on the cell type (normal vs. early cancer cells), as determined in short-term experiments. Most importantly, we have observed increases in NTR1 expression in irradiated normal prostate epithelial cells, validating our hypothesis that radiation-induced cancer transformation could be accompanied by NTR1 activation. In addition, we have demonstrated the crucial role of NTR1 in radiation-induced signal transduction through EGFR-SRC-STAT5 pathway. In addition to the experimental work, we have completed the assembling of our research team. Mr. Nicholas Valerie graduated from NASA Space Radiation Summer School, and participated in NSRL10B and 10C experiments. Dr. Suresh Anaganti also fulfilled all the training requirements and participated in NSRL10C experiment.
Bibliography Type:	Description: (Last Updated: 11/14/2018)
Abstracts for Journals and Proceedings	Dziegielewski J, Valerie NCK, Hosing AS, Casarez EV Jr., Parsons SJ, Larner JM. "Effects of prolonged exposure to space radiation on carcinogenesis and neuroendocrine differentiation in human prostate cells." Presented at the NASA Human Research Program Investigators' Workshop, Houston, TX, February 2010. Abstract book. NASA Human Research Program Investigators' Workshop, Houston, TX, February 2010. , Feb-2010
Abstracts for Journals and Proceedings	Dziegielewski J, Valerie NCK, Parsons SJ, Larner JM. "Carcinogenic transformation and neuroendocrine differentiation induced by space radiation in human prostate cells." Presented at the 21st NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 2010. Abstract book. 21st NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 2010. , May-2010
Abstracts for Journals and Proceedings	Dziegielewski J, Valerie NCK, Dasilva JO, Hosing AS, Parsons SJ, Larner JM. "Interplay between neurotensin receptor 1 (NTR1) and epidermal growth factor receptor (EGFR) in responses to ionizing radiation in prostate cells." Presented at the 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. Abstract book. 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=1bbdf105-ce87-4ade-9ec5-faa9dcf52215&cKey=53757af6-8caa-4fe8-9222-0e9df1e169c2&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Articles in Peer-reviewed Journals	Dziegielewski J, Goetz W, Baulch JE. "Heavy ions, radioprotectors and genomic instability: implications for human space exploration." Radiat Environ Biophys. 2010 Aug;49(3):303-16. PMID: 20035342 , Aug-2010